

Product: a whole prescription

Dolo 650 × 30, Stamlo 5 and Atorbest 10, delivered to PIN 110001.

Your prescription

One medicine per line, e.g. "Dolo 650 x30". We find the cheapest way to buy all of it.

COMMON PRESCRIPTIONS

DiabetesHigh blood pressureCold, cough & feverAcidityCholesterolThyroid

MEDICINE (BRAND OR SALT WITH STRENGTH)

TABLETS (OPTIONAL)

Dolo 650

30

X

Stamlo 5

1 pack

X

Atorbest 10

1 pack

X

+ Add medicine

Try an example

110001

Find the cheapest basket

Up to 8 medicines. Leave the count empty to buy one pack (or one bottle/tube). Every medicine is searched at the same time.

1

Cheapest overall

BEST VALUE

Same-salt swaps, from any pharmacies

₹138.39

1 order: SastaSundar · ₹29.00 delivery

SastaSundar

₹29.00 delivery

₹109.39 + ₹29.00 = ₹138.39

Dolo 650

Micro Labs Ltd · 2 packs × ~15 tablets

₹52.68

~₹1.76/tablet

Buy

Amodep 5

SWAP for Stamlo 5

FDC Ltd · 2 packs × 15 tablets

₹30.16

₹1.01/tablet

Buy

Atorbest 10

Cadila Pharmaceuticals Ltd · 1 pack × ~10 tablets

₹26.55

~₹2.66/tablet

Buy

Open Dolo 650 (1 of 3)

Copy shopping list

2

One pharmacy

Same-salt swaps, everything in one order

₹138.39

1 order: SastaSundar · ₹29.00 delivery

SastaSundar

₹29.00 delivery

₹109.39 + ₹29.00 = ₹138.39

Dolo 650

Micro Labs Ltd · 2 packs × ~15 tablets

₹52.68

~₹1.76/tablet

Buy

Amodep 5

SWAP for Stamlo 5

FDC Ltd · 2 packs × 15 tablets

₹30.16

₹1.01/tablet

Buy

Atorbest 10

Cadila Pharmaceuticals Ltd · 1 pack × ~10 tablets

₹26.55

~₹2.66/tablet

Buy

Open Dolo 650 (1 of 3)

Copy shopping list

3

Exactly as prescribed

Your brands, at their cheapest pharmacies

₹174.25

+₹35.86 more

1 order: SastaSundar · ₹29.00 delivery

SastaSundar

₹29.00 delivery

₹145.25 + ₹29.00 = ₹174.25

Dolo 650

Micro Labs Ltd · 2 packs × ~15 tablets

₹52.68

~₹1.76/tablet

Buy

Stamlo 5

Dr Reddy's Laboratories Ltd · 1 pack × 30 tablets

₹66.02

₹2.20/tablet

Buy

Atorbest 10

Cadila Pharmaceuticals Ltd · 1 pack × ~10 tablets

₹26.55

~₹2.66/tablet

Buy

Open Dolo 650 (1 of 3)

Copy shopping list

4

Medicine by medicine

Try place to buy each medicine, cheapest per tablet first (delivery included when bought on its own).

Dolo 650

30 tablets · Paracetamol 650mg tablet · 18 offers

From ~₹3.69/tablet

#	PHARMACY · PRODUCT	PER TABLET	DELIVERY	YOU PAY
1	SastaSundar CHEAPEST Micro Labs Ltd · Dolo 650 mg	~₹3.69	₹29 delivery · add ₹122.66 more for FREE delivery 3 Days	₹55.34 Visit site
2	Dolopar 650 at SastaSundar SAME SALT Micro Labs Ltd · Dolopar 650 mg	~₹3.69	₹29 delivery · add ₹122.66 more for FREE delivery 3 Days	₹55.34 Visit site
3	Pacimol 650 at SastaSundar SAME SALT Ipca Laboratories Ltd · Pacimol 650 mg	~₹3.70	₹29 delivery · add ₹122.52 more for FREE delivery 3 Days	₹55.48 Visit site
4	Paracip 650 at Dawaa Dost SAME SALT Cipla Ltd · Cipla Paracip 650mg Pain Relief Tablets 15s	₹4.93	₹50 delivery · add ₹826 more for FREE delivery 1-2 Days	₹74.00 Visit site
5	1mg Micro Labs Ltd · Dolo 650 Tablet	~₹5.22	₹50 delivery · add ₹471.70 more for FREE delivery 1-2 Days	₹78.30 Visit site
6	Dawaa Dost Micro Labs Ltd · Micro Labs Dolo 650mg Tablets 15s	~₹5.27	₹50 delivery · add ₹821 more for FREE delivery 1-2 Days	₹79.00 Visit site
7	Calpol 650mg at 1mg SAME SALT Gloxo SmithKline Pharmaceuticals Ltd · Calpol 650 Tablet	~₹5.36	₹50 delivery · add ₹469.60 more for FREE delivery 1-2 Days	₹80.40 Visit site
8	Medizinhub Micro Labs Ltd · Dolo 650 Mg Tablet	~₹5.40	₹50 delivery · add ₹1,969 more for FREE delivery 1-2 Days	₹81.00 Visit site
9	Netmeds Micro Labs Ltd · Dolo 650 Tablet 15s	₹5.61	₹50 delivery · add ₹474.82 more for FREE delivery 1-2 Days	₹84.18 Visit site
10	P 650 at 1mg SAME SALT Apex Laboratories Pvt Ltd · Buy P 650 Tablet Online: View Uses, Side Effects, Price...	~₹7.06	₹50 delivery · add ₹476.40 more for FREE delivery 1-2 Days	₹70.60 Visit site

More options · show 8 more

Stamlo 5

30 tablets · Amlodipine 5mg tablet · 15 offers

From ₹2.94/tablet

#	PHARMACY · PRODUCT	PER TABLET	DELIVERY	YOU PAY
1	Amodep 5 at SastaSundar CHEAPEST FDC Ltd · Amodep 5 mg Tablet (15 Tab)	₹2.94	₹29 delivery · add ₹133.92 more for FREE delivery 3 Days	₹44.08 Visit site
2	SastaSundar Dr Reddy's Laboratories Ltd · Stamlo 5 mg Tablet (30 Tab)	₹3.17	₹29 delivery · add ₹92.98 more for FREE delivery 3 Days	₹95.02 Visit site
3	Amlopres 5 at SastaSundar SAME SALT Cipla Ltd · Amlopres 5 mg Tablet (30 Tab)	₹3.17	₹29 delivery · add ₹92.89 more for FREE delivery 3 Days	₹95.11 Visit site
4	Amlodac 5 at Dawaa Dost SAME SALT Zydus Cadila · Zydus Cadila Amlodac 5mg Tablets 30s for...	₹3.97	₹50 delivery · add ₹781 more for FREE delivery 1-2 Days	₹119.00 Visit site
5	1mg Dr Reddy's Laboratories Ltd · Stamlo 5 Tablet	~₹4.00	₹50 delivery · add ₹429.90 more for FREE delivery 1-2 Days	₹120.10 Visit site
6	Adp 5mg at SastaSundar SAME SALT Capet India Pvt Ltd · Adp 5 mg Tablet (15 Tab)	₹4.13	₹29 delivery · add ₹115.99 more for FREE delivery 3 Days	₹62.01 Visit site
7	Chemist180 Dr Reddy's Laboratories Ltd · Stamlo 5MG Tablet	~₹5.53	₹100 delivery · add ₹933.98 more for FREE delivery 2-4 Days	₹166.02 Visit site
8	Amlong at Chemist180 SAME SALT Micro Labs Ltd · Amlong 5MG Tablet	~₹5.53	₹100 delivery · add ₹933.98 more for FREE delivery 2-4 Days	₹166.02 Visit site
9	PharmEasy Dr Reddy's Laboratories Ltd · Stamlo 5Mg Strip Of 30 Tablets	₹6.83	₹100 delivery · add ₹868.01 more for FREE delivery · +₹10 platform fee 1-2 Days	₹204.99 Visit site
10	Amlopin 5 at Dawaa Dost SAME SALT USV Ltd · USV B1812'S Amlopin 5mg Tablets 10s for Blood...	₹7.30	₹50 delivery · add ₹927 more for FREE delivery 1-2 Days	₹73.00 Visit site

More options · show 5 more

Three ways to buy, delivery included

1

Cheapest overall

₹138.39

Amodep 5 swap · 1 order: SastaSundar · ₹29 delivery

2

One pharmacy

₹138.39

The same single SastaSundar order

3

Exactly as prescribed

₹174.25

Prescribed brands · 1 order: SastaSundar

4

Medicine by medicine

Every pharmacy and same-salt brand for each line, cheapest per tablet first.

One same-salt swap saves ₹35.86 (21%)

Whole packs: Dolo 650 as 2 × ~15, Amodep 5 as 2 × 15, Atorbest 10 as 1 × ~10. 1,159 basket combinations priced.

Captured 9 Oct 2026 · delivery fees from configured pharmacy rules

How the engine works behind the scenes

What runs between a search and an answer, and the decisions that keep SerpApi use low.

1 • From a name to a medicine, 0 credits

- Every query is looked up first in a SQLite catalogue of 246,046 Indian medicines and 17,028 compositions.
- A name without a strength ("paracetamol") returns the options instead of searching; the user picks one.
- Misspellings are corrected against the catalogue ("dollo 650" → Dolo 650). Gemini only chooses among candidates the catalogue supplies, and its pick is cached for 30 days.

2 • One main search, prices in about 3 seconds

- One SerpApi google_shopping call; json_restrictor returns only the listing fields we use.
- Listings are mapped to 10 pharmacies, and each title must match the brand and strength (look-alikes such as Stamlo Bis are rejected).
- Delivery and platform fees for the PIN are added, and the main list streams to the browser over SSE while the rest of the search continues.

3 • Few results? Widen the pool

- Google often returns few or none of the searched brand: "Pan 40 price" gave 24 listings and not one Pan 40.
- So up to three same-salt searches run in parallel (salt + "tablet generic", the same + a pharmacy's name, a discounted-generic brand), plus a "brand + tablet" retry when the brand is missing.
- Every listing of every search is matched against every brand of the composition (281 for Stamlo 5), so the searched medicine is also priced from the alternatives' results.
- Thin results (fewer than 3 real listings) are cached for 1 hour, not 24, so they refresh sooner.

4 • Keeping SerpApi cost low

- A repeat within 24 h is an exact Redis hit: 0 credits, 7.6 ms on average against 12.9 s for a SerpApi call (measured 9 Oct 2026, 8 hits and 17 calls).
- Similar wording can reuse a cached search (MiniLM cosine ≥ 0.88), but only with the same parameters and the same numbers.
- Product pages are looked up front only for the cheapest offer and the cheapest alternative (8 → 2 per new medicine); any other page costs 1 credit when clicked, then is cached 24 h.

5 • What one medicine costs

- New medicine on the website: typically 4 credits (3 for a combination), +1 for the brand retry, up to 8 with product pages.
 - In a prescription or through MCP: the same searches, with a product page only for the offer the user picks.
 - Repeat within 24 h: 0.
- A name without a strength: 0.
- Guards: a daily budget (30 credits on the hosted API), at most 3 searches at once, and an optional second key used only after the first runs out.

6 • Other decisions that pay off

- Everything after the catalogue lookup runs in parallel: in the Telma 40 run, 9 calls took 33.8 s one after another, but main results arrived at 2.6 s and alternatives at 5.2 s.
- The basket is optimised with branch-and-bound: exact over the cheapest offer per pharmacy and medicine, and checked against brute force on 200 random baskets.
- The website, the API and the MCP tools call the same Python functions, so they share one cache, one credit budget and the same answers.

PART 2

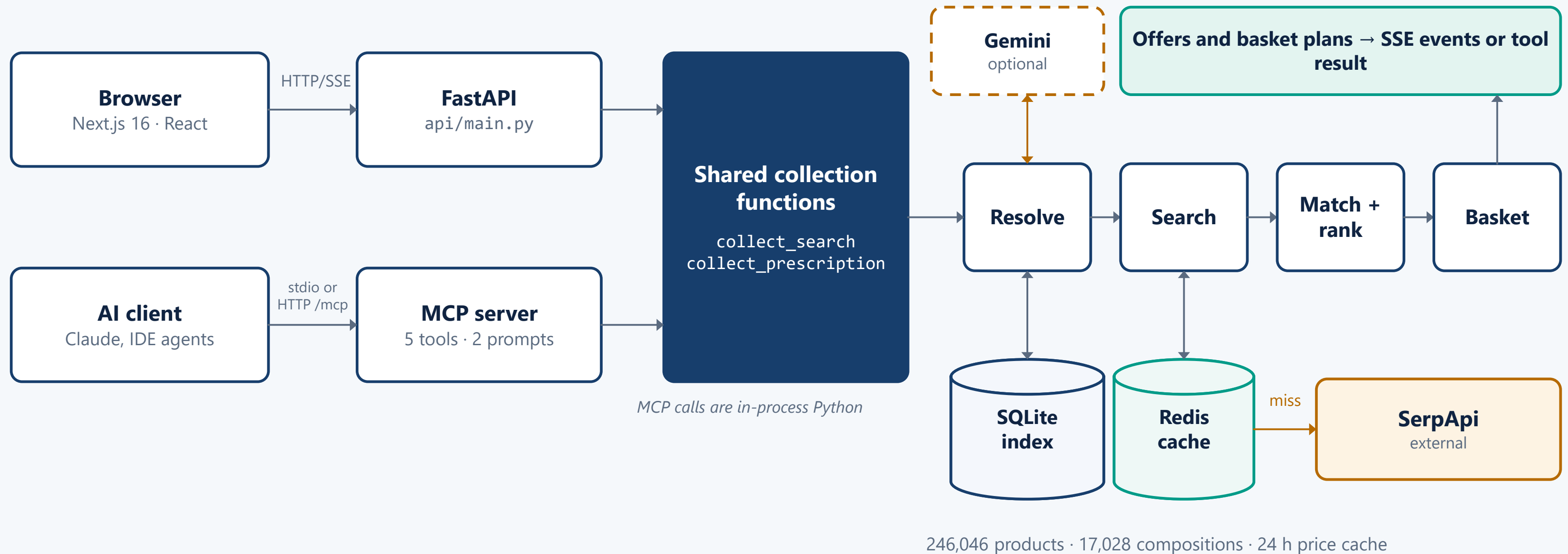
The engineering

How search listings become delivered prices and purchase plans.



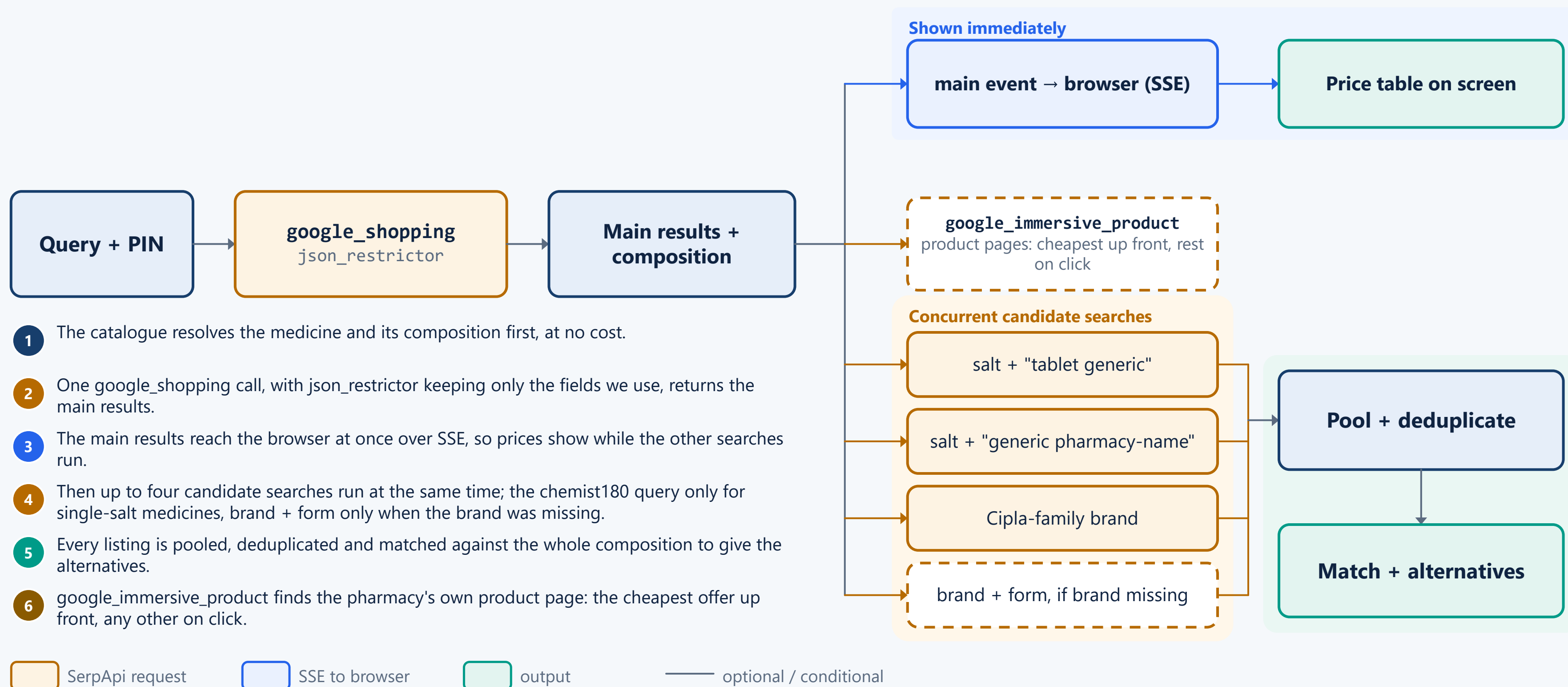
One backend, browser and agent access

A browser and an AI client call the same Python functions and pipeline.



SerpApi searches expand the offer pool

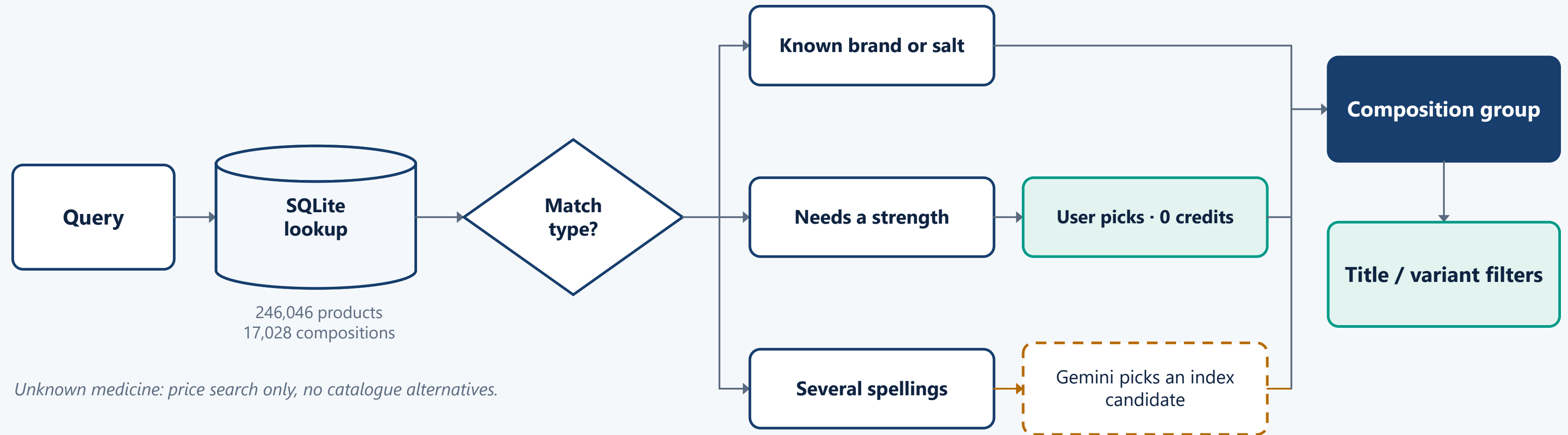
google_shopping finds listings. json_restrictor trims fields. google_immersive_product finds product pages.



- 1 The catalogue resolves the medicine and its composition first, at no cost.
- 2 One google_shopping call, with json_restrictor keeping only the fields we use, returns the main results.
- 3 The main results reach the browser at once over SSE, so prices show while the other searches run.
- 4 Then up to four candidate searches run at the same time; the chemist180 query only for single-salt medicines, brand + form only when the brand was missing.
- 5 Every listing is pooled, deduplicated and matched against the whole composition to give the alternatives.
- 6 google_immersive_product finds the pharmacy's own product page: the cheapest offer up front, any other on click.

Catalogue-grounded medicine matching

Identity comes from a SQLite catalogue. Gemini only picks among candidates the index supplies.



COMPOSITION KEY

salt(s) + strength(s) + form + release

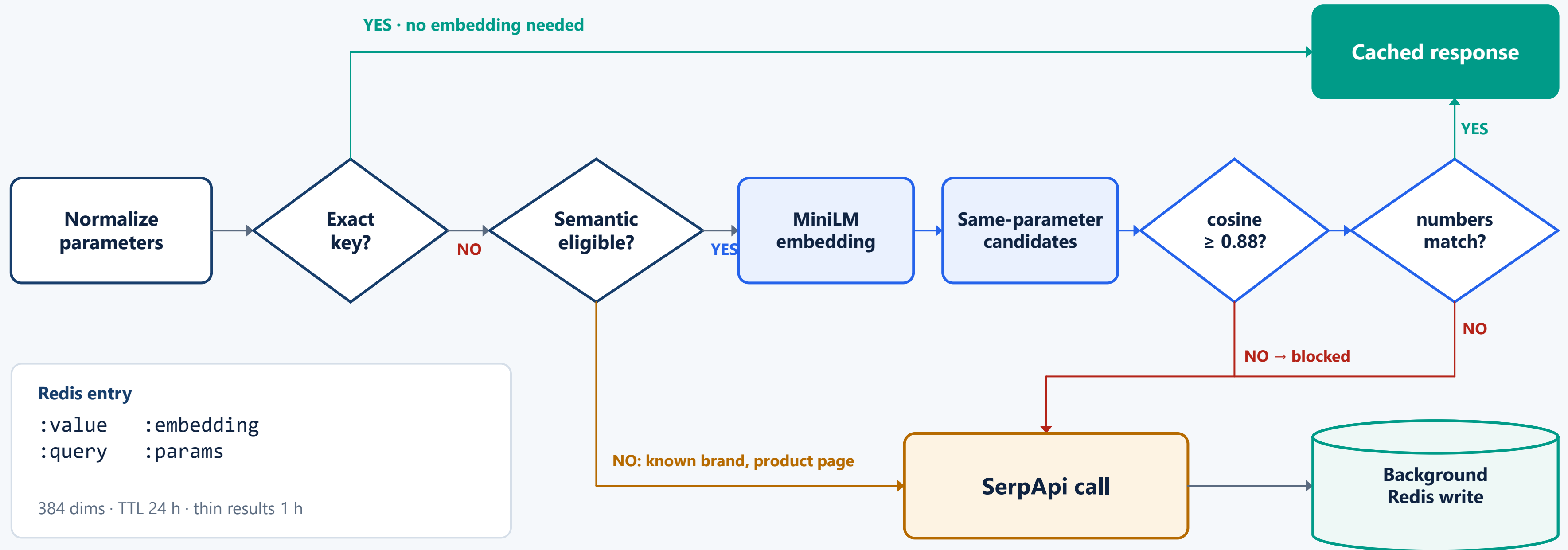
Stamlo 5 → amlodipine:5mg|tablet| (no release modifier)

Stamlo 5 listing ✓ kept

Stamlo Bis 5 listing ✗ rejected: different product

Cache reuse with explicit constraints

Exact keys first. Semantic reuse only where it is safe, then parameter and numeric checks.



Dolo 500 vs Dolo 650: cosine 0.949, numbers differ → SerpApi, not a cached Dolo 500

Search listings become delivered estimates

What a listing says, what our delivery rules add, and the price you pay.

$$\text{Delivered estimate} = \text{item subtotal} + \text{delivery fee} + \text{platform fee}$$

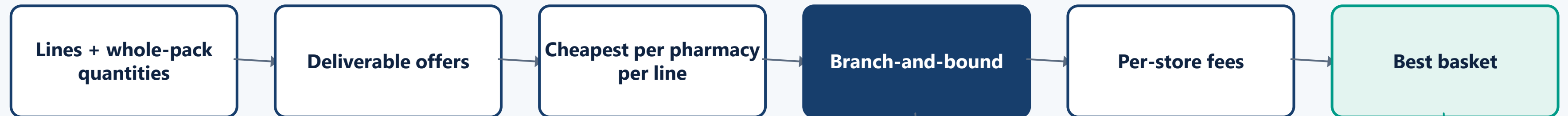
Field	Comes from	Apollo · Stamlo-5 15's · PIN 110001
Pharmacy	Search: seller name	Apollo Pharmacy
Product and pack	Search: title, checked against the catalogue	Stamlo 5, 15 tablets
Item price	Search: listing price	₹40.00
PIN zone	Rules: PIN prefix	110001 → metro
Delivery fee	Rules: pharmacy, zone and order subtotal	₹93.22 (₹79 + GST, order below ₹199)
Platform fee	Rules: pharmacy and order subtotal	₹0 below ₹199
Delivered estimate	Sum of the above	₹133.22

WHAT THE PIPELINE DOES

- 1 SerpApi's Google Shopping search returns the seller and the listing price.
- 2 The seller is matched to one of 10 pharmacies; the title is checked against the medicine, and duplicates are dropped.
- 3 Your PIN maps to a delivery zone: metro, tier 2, tier 3, remote or unserviceable.
- 4 That pharmacy's rules add its delivery fee, free-delivery threshold and platform fee on the order subtotal.
- 5 Fees come from checkout carts or the pharmacy's Google product page. Medplus publishes none, so it is shown but never picked as cheapest.

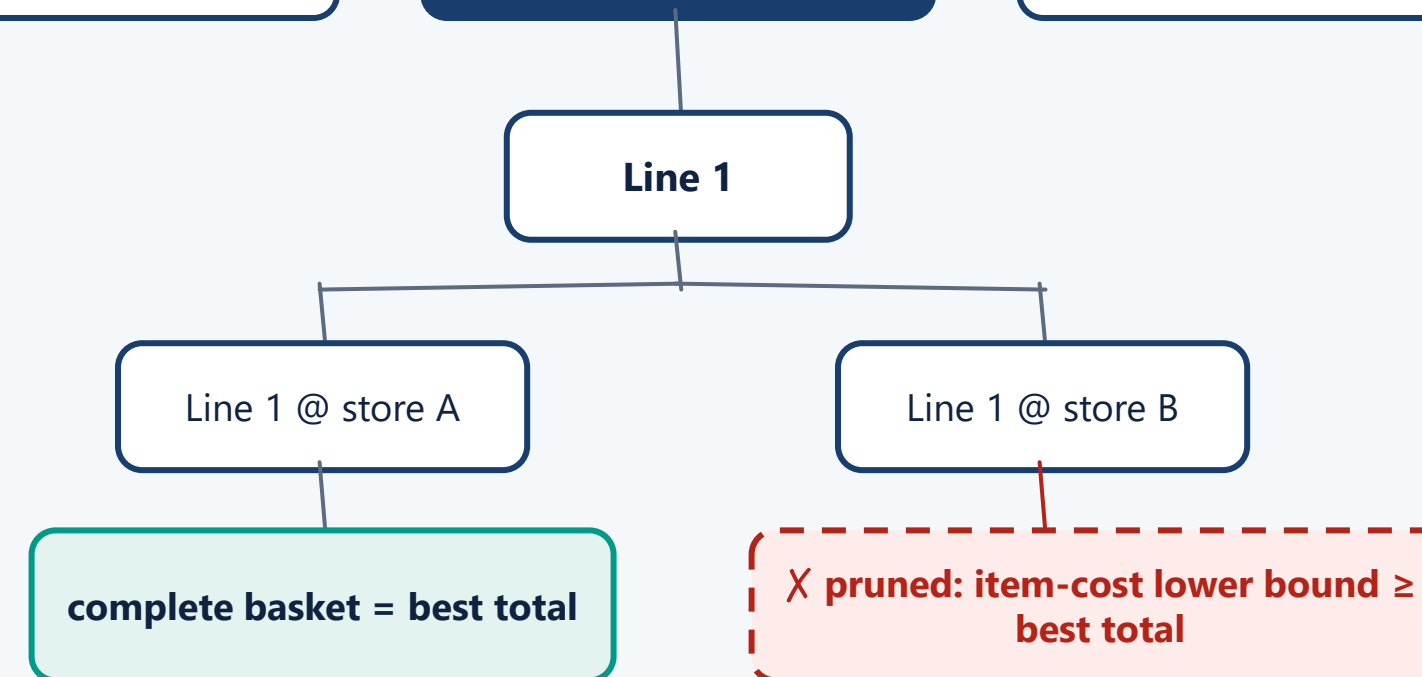
Whole-prescription optimisation

Fees apply to each pharmacy's subtotal, so the cheapest items can still lose.

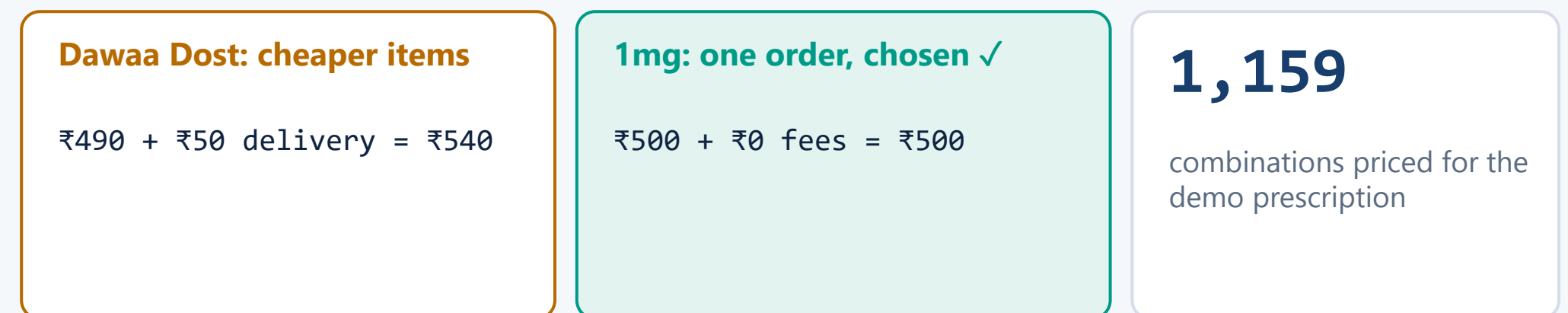


How the basket is chosen

- 1 Each medicine becomes whole packs.
- 2 Each pharmacy keeps its cheapest offer per medicine.
- 3 Branch-and-bound cuts a branch once its item cost alone reaches the best total.
- 4 Fees are added on each pharmacy's subtotal.
- 5 Three plans come out.



TEST FIXTURE · synthetic offers, configured delivery rules

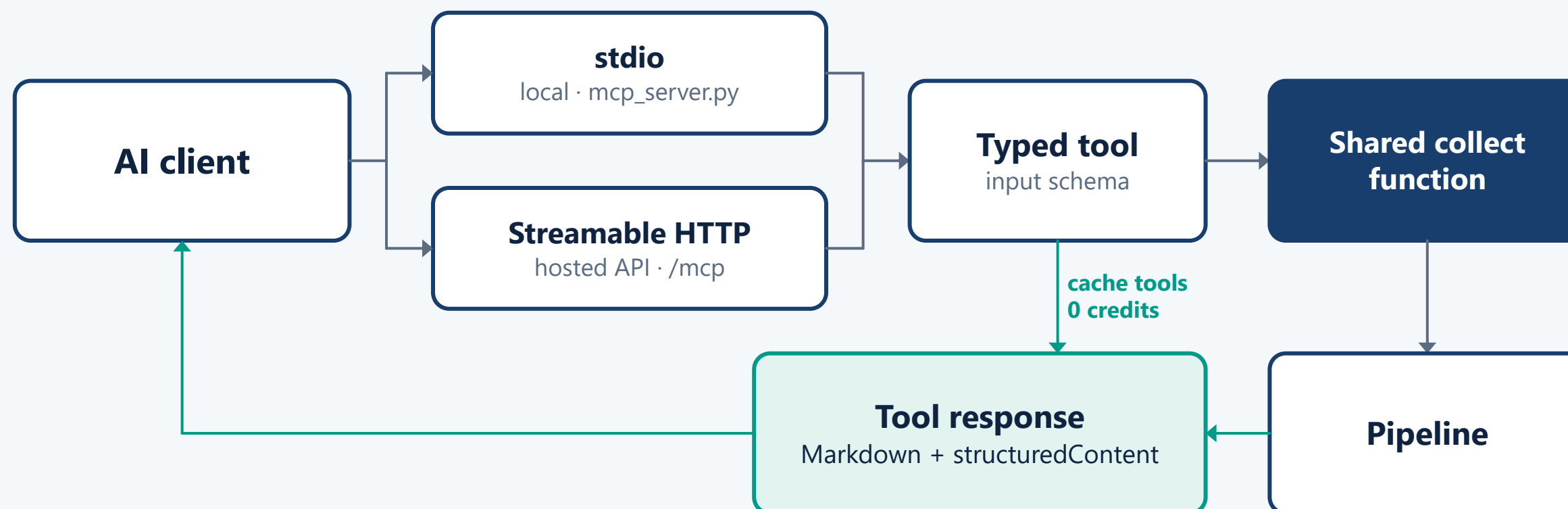


Up to 8 medicines per prescription · node cap 2,000,000

MCP tools make the workflow agent-accessible

Typed tools over stdio or Streamable HTTP, answering in Markdown plus structured content.

CORE FEATURE



Why MCP: an assistant cannot see delivery fees or same-salt brands from a web search. MCP gives any client (Claude, Cursor, IDE agents) the same engine, cache and credit budget as the website, so an agent gets the answers a shopper gets.

5 TOOLS · 2 PROMPTS

search_medicine

delivered offers + same-salt swaps · 3–5 credits new, 0 cached

plan_prescription

up to 8 lines, three basket plans

get_buy_link

pharmacy product page · 1 credit, then 0 for 24 h

cache_lab · cache_stats

inspect the cache · 0 credits

prompts

compare_medicine · plan_my_prescription

BUILT FOR AGENTS

- Typed inputs; every tool publishes an outputSchema and returns structuredContent plus Markdown
- listing_id on every offer → get_buy_link only for the pick
- Strength prompt and stated spelling fixes, 0 credits
- Coded tool errors before any credit is spent
- Progress notifications; read-only annotations; logs on stderr
- Stateless Streamable HTTP at /mcp, or local stdio

Real call · 9 Oct 2026

```
plan_prescription(
  medicines=[Dolo 650 x30,
             Stamlo 5, Atorbest 10],
  pincode="110001")
```

Cheapest overall: ₹138.39

₹109.39 + ₹29.00 delivery

1 order: SastaSundar

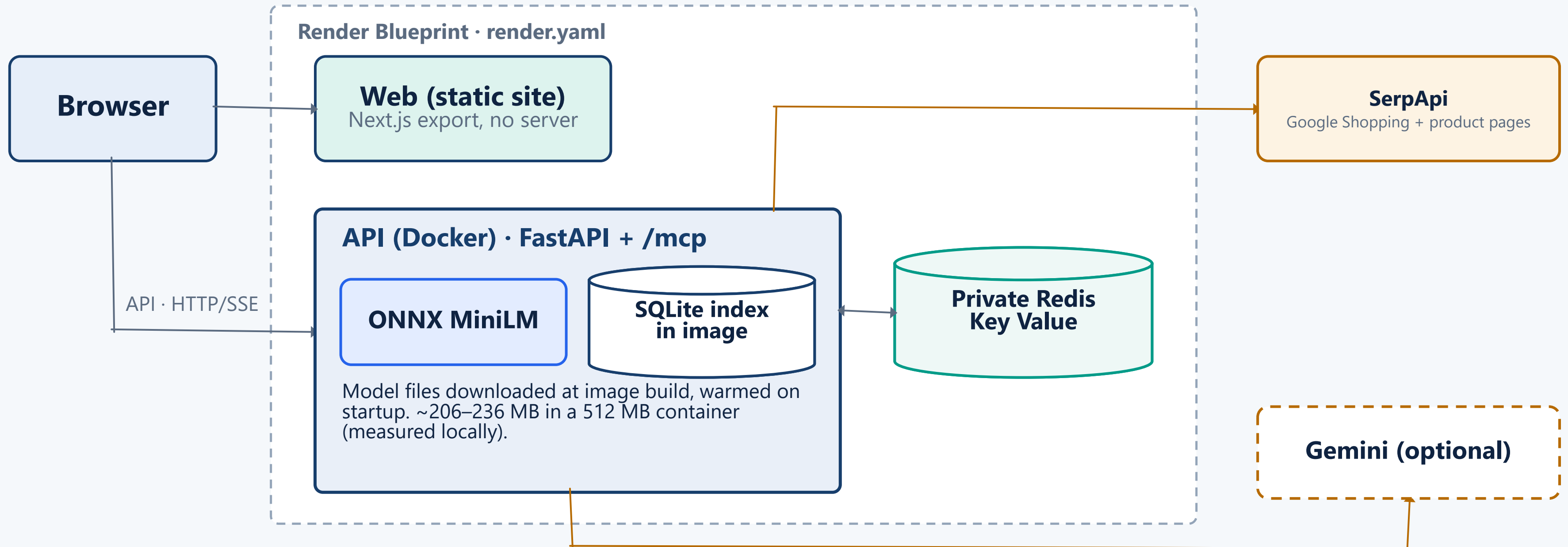
Dolo 650	2 × ~15	₹52.68
Amodip 5	2 × 15	₹30.16
Atorbest 10	1 × ~10	₹26.55

As prescribed: ₹174.25

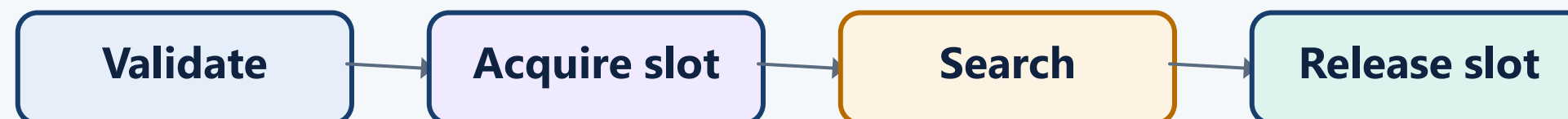
11 lookups · 11 from cache
0 credits

Deployment and controlled resource use

A static site, one Docker API and a private Redis on Render's free plan, with request and credit limits.



EVERY SEARCH REQUEST



Blueprint config: 30 credits/day · 3 concurrent searches. Keep-awake: GitHub Actions calls /api/health every 10 min

Data we can stand behind

Every fee has a source, every swap has a reason, and nothing is guessed.

Where the data comes from

Medicine catalogue

Indian Medicine Dataset (JuniorAlive, MIT licence), 253,973 rows, rebuilt into 246,046 products, 17,028 compositions and 7,641 manufacturers by `scripts/build_medicine_index.py`.

Prices

Live Google Shopping results through SerpApi at search time, cached for at most 24 hours. No price is typed in by hand.

Delivery fees

One rules file, `notes/postal_codes_delivery_rules.json`. Every pharmacy entry names its source and date.

Checks

44 rules-integrity checks (every pharmacy recognised and priced), 37 delivery-fee checks and 126 matching checks pass (9 Oct 2026).

Delivery fees

Checkout carts

Apollo (PIN 110001), PharmEasy (250002), 1mg (Gurugram), Truemeds (110044): fees read from real carts, Oct 2026.

Google product-page shipping

Netmeds, SastaSundar, Chemist180, Dawaa Dost (SerpApi `google_immersive_product`, 3 Oct 2026; Gurugram and Guwahati identical) and Medizinhub (9 Oct 2026).

No published fee

Medplus: shown, never picked as cheapest.

Removed, fees unverifiable

Kogland, Medivik, Magicine, eMedicalwala: removed rather than priced on a guess.

Safe by design

Swaps need an identical composition key

Salt, strength, form and release must all match; listing titles are checked against it, so look-alikes are rejected.

Every swap carries a warning

Ask a doctor or pharmacist before switching. Not medical advice.

Estimates are labelled

Estimated pack sizes are marked ~; unknown fees stay visible in red.

Keys and credits are protected

Keys never in code or images; daily budget and concurrency limits.

Problems we hit, and how we fixed them

Six of the eight documented in the README, and why each fix is necessary.

A semantic cache confused strengths

Problem "Dolo 500" and "Dolo 650" score 0.949 cosine.

Fix A numeric guard; brand searches are looked up by exact name only.

Why a wrong cache hit shows another medicine's prices.

Titles often omit pack size

Problem "Stamlo 5MG Tablet ₹66.02": 15 or 30 tablets?

Fix Estimate from the brand's other listings, then the catalogue; mark it ~.

Why without a per-tablet price, the cheapest offer was skipped.

Google returns look-alike medicines

Problem A Stamlo 5 search: 22 listings, only 8 Stamlo 5 (others: Esta 5, Stamlo Bis).

Fix Titles must match brand and strength; variants and second doses are rejected.

Why comparing a different drug is wrong and unsafe.

Cheapest generics are cheap only after discount

Problem Pharmacies' suggestions ranked 449th of 500 by list price.

Fix Extra searches for reach ("tablet generic", a pharmacy name, a discounted brand), pooled.

Why otherwise we lose to the pharmacy's own suggestion.

A brand search often misses the brand

Problem "Pan 40 price": 24 listings, none of Pan 40.

Fix Pool every search against the whole composition; retry with "Pan 40 tablet".

Why a 5-medicine basket went from 3 compared medicines to 5.

Step by step was slow

Problem 9 SerpApi calls in a row took 33.8 s.

Fix Run in parallel and stream: main results at 2.6 s, alternatives at 5.2 s.

Why shoppers see prices in seconds.

PART 3

What comes next

Making the data stronger, with doctors and pharmacists.



Where we take it next

From a working hackathon build to a service patients and doctors can rely on.

Built and tested

- Delivered price for 10 pharmacies, 9 with verified fees
- Same-composition swaps from 246,046 products
- Three basket plans for a prescription
- 5 MCP tools, hosted and local
- Every SerpApi call visible and cached

TEST RESULTS AND MEASURED PERFORMANCE · 9 OCT 2026

200 / 200 random baskets: branch-and-bound total = brute-force total, 0 mismatches

Automated checks, all passing: 62 API · 92 MCP · 126 matching + pipeline · 25 basket · 37 delivery fees · 44 rules integrity

Pan 40, first run: 19.2 s · 4 credits Repeat: 111 ms · 0 credits

Prescription from MCP, cached: 11 lookups · 0 credits

Basket search: 1,159 combinations

Next

- **Catalogue reviewed with doctors and pharmacists:** clinicians confirm which brands are interchangeable.
- **Regular data refresh, full multi-salt compositions:** new brands recognised; today's dataset stores at most two salts.
- **Verified delivery fees for every pharmacy,** and more pharmacies.
- **Publish the connector to the MCP Registry:** server.json is ready.

Later

- **Prescription photo to basket** (OCR).
- **Price alerts** for monthly refills.
- **Clinician-reviewed lists** of interchangeable brands.

The catalogue decides what counts as an alternative, so every improvement to it makes every search stronger.

Team PharmaWatch

SerpApi India Hackathon 2026 · Track: Commerce & Market Intelligence



Jatin Singh

Team PharmaWatch



Kush Pathak

Team PharmaWatch



Scan for the repo

PharmaWatch turns search listings into medicine and prescription purchase plans.

github.com/b25cs1051-KUSH/Serp-API-Hackathon